

MILAD RABIEI

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🌐 milwd.github.io

📖 [Google Scholar](#)

Education

Master of Science, Robotics Engineering

2024 - Present

DIBRIS, Università degli Studi di Genova, Italy

Thesis (undergoing): Intra-task semantic generalization from motion constraints

Advisors: [Elena Merlo](#), [Marta Lagomarsino](#), and [Arash Ajoudani](#)

Bachelor of Science, Electrical Engineering

2019 - 24

ECE Dept., Shahid Beheshti University, Iran

Thesis: Landmark tracking and object detection, complete vision system for self-driving cars

Advisor: [Mohammad Hossein Moaiyeri](#)

Publications

Mutual Impact of Feature Selection and Privacy-preserving Mechanisms

International Journal of Machine Learning and Cybernetics, [Springer](#), 2025

M Alishahi, V Moghtadaiee, A Fathalizadeh, Milad Rabiei

Differentially Private GANs for Generating Synthetic Indoor Location Data

International Journal of Information Security, [Springer](#), 2025

V Moghtadaiee, M Alishahi, Milad Rabiei

A Density-based Lane Keeping System for A Cost-efficient Autonomous Vehicle Platform

Manuscript, [Arxiv](#), 2024

Younesi*, F., Milad Rabiei*, Keivanfard*, S., Sharifi*, M., GhayourNajafabadi, M., Moadeli, B., Jafari, A., Moaiyeri, M. H.

Academic Experiences

Guest Student - [Human-robot interfaces and interaction \(HRI²\)](#), IIT

2026-

- Thesis work on neurosymbolic robot manipulation and imitation learning

Research Assistant - [Cyberspace research institute](#), SBU

2023 - 24

- Generative networks, private data analysis, differential privacy, federated learning

Research Assistant - [Robotics and intelligent automation lab](#), SBU

2020 - 23

- Vision for AGVs (level 4), segmentation, detection, and localization

Lead Teaching Assistant - ECE department, SBU

2021 - 24

- Machine Learning (graduate-level), Prof.: [Reza Ghaderi](#)
- Intro to Artificial Intelligence (graduate-level), Prof.: [Atefeh Aghaei](#)
- Computer Programming, Prof.: [Vahideh Moghtadaiee](#)

Skills

Programming/Frameworks: Python (OpenCV, PyTorch, TF/Keras, etc.), C/C++, MATLAB, Mujoco, PDDL/PDDL+, FastDDS, ROS/ROS 2, Gazebo, SAM, VLMs (Llama3, OpenAI API)

Embedded: Nvidia Jetson (CUDA), Raspberry Pi, dsPIC, NodeMCU ESP8266, ESP32

Languages: English (proficient), Italian (elementary), Persian (native)

Honors and Awards

(2023) Recipient of MCI R&D research grant for small-size autonomous vehicle
(2022) Recipient of honorary Dean and Departmental research grants for Robotics Lab at SBU
(2022) 1st place, Individual team section, Autonomous vehicles league, IranOpen RoboCup 2022
(2022) 1st place, Technical challenge, Autonomous vehicles league, IranOpen RoboCup 2022
(2021) 2nd place, Race section, Autonomous cars simulation, Fira RoboWorld Cup SDE
(2021) 1st place, Race section, Autonomous cars simulation, Iran Fira RoboWorld Cup
(2021) 2nd place, Urban section, Autonomous cars simulation, Iran Fira RoboWorld Cup
(2019) Top 99th percentile in national university entrance exam among 140K participants

Other Experiences

(2026-) Junior reviewer ([Young Reviewers Program](#)), IEEE Robotics and Automation Society

Conference Coordinator:

(2025) The 19th International Conference on Intelligent Autonomous Systems ([IAS19](#)), Genoa, Italy.
(2022) The 4th Iranian International Conference on Microelectronics ([IICM](#)), Tehran, Iran
(2022) The 13th Power Electronics and Drives: Systems and Technologies Conference ([PEDSTC](#)), Tehran, Iran

(2024) Member of National Committee, [IranOpen RoboCup International Competitions](#)

(2023) Member of Technical Committee, [IranOpen RoboCup International Competitions](#)

(2024) Python Programming Course Instructor, [Danesh High School](#), Tehran, Iran

(2023) Machine Learning Course Instructor, [Iran RoboCamp](#), Tehran

(2023) Organizer of Deep Learning Reading Group, SBU ECE Department

(2022) Network Programming Intern, Amin Optical Communications Technologies (FANA), Tehran

Tools and Projects

Few-shot gaussian splatting with diffusion-based data augmentation

[Warehouse Automation with PDDL+](#)

[Drone Simulation with Remote Access \(FastDDS\)](#)

[Q-Learning for Cartpole in Gym](#)

[Task Priority Kinematic Control for Mobile Manipulation and Bimanipulation](#)

[1:10 Scale Self-driving Car](#)

[Ball Balancer with PID](#)

[Analysis of private neural data generation \(DPGANs\)](#)

[Object detection dataset synthesizer](#)

[\(Mediapipe\) Hand gesture recognition and ESP32 control](#)

[GAN/conditional GAN for MNIST dataset](#)

[Image recovery with genetic algorithm](#)

[Sobel edge detection in VHDL](#)